

Parameter	Symbol	Formula	Calculation	Estimated Value	Unit	Description
Frontal Area	A	$A = W \times H \times k$	-	-	m ²	Estimated projected frontal area
Drag Coefficient	Cd	SEM Regulation	-	70	kg	Minimum driver weight according to Shell Eco-marathon regulations
Drag Force	Fd	$m = m_{\text{vehicle}} + m_{\text{driver}}$	85 + 70	155	kg	Total mass used in vehicle dynamics calculations
Length	L	Standard atmosphere	-	1.225	kg/m ³	Air density at sea level
Height	H	Assumption	-	0.1	-	Estimated aerodynamic drag coefficient of a competitive SEM Urban Concept vehicle
Front Track Width	W	Assumption	-	1.22	m	Estimated within SEM regulations
Rear Track Width	H	Assumption	-	1.05	m	Estimated within SEM regulations
Wheel Base	k	Assumption	-	0.6	-	Correction factor for streamlined body shape
Ground Clear	A	$A = W \times H \times k$	$1.22 \times 1.05 \times 0.60$	0.77	m ²	Estimated projected frontal area
Vehicle Weight	Crr	Assumption	-	0.004	-	Typical value for low rolling resistance SEM tyres

[illegible]